Final Review

Ice Breakers
Lincoln County High School
Team #3

THE UNIVERSITY OF
ALABAMA IN HUNTSVILLE

inSPIRESS
The Innovative System Project for the Increased Recruitment of Emerging STEM Students

ICE
BREAKERS

Agenda

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  - UAH Baseline Mission
- Defining the Problem
  - Potential Science Objectives
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  - Objective Chosen
  - Science Traceability Matrix
  - Instrumentation
  - Payload Design Requirements
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- Payload Concepts
  - Concepts 1 and 2
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  - Chosen Concept
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- Engineering Analysis
  - Initial Conditions
  - Deployment
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  - Final Design
  - Team's Mission
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  - Activities for Community Engagement (ACEs)
  - Online Outreach
- Closing Remarks
  - Lessons Learned
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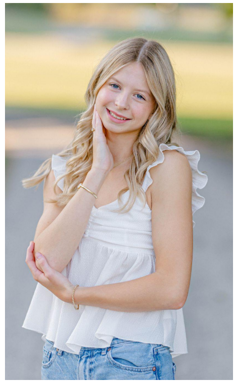
ICE BREAKERS

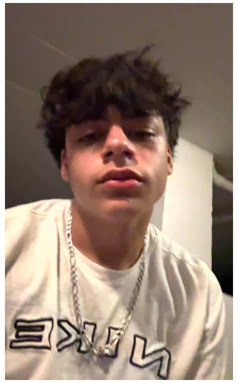
Team

Project Manager
Alivia Phelps



CAD Lead
Quin Gordon

Outreach Lead
Didier Aguilar



Design Lead
Brody Huskey

CAD team
Travis Griffin

Outreach team
Lawton Norton

Design team
Colten Osborn

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Team Identity

Ice Breakers

The image of the icy crater represents our mission to research the ice in the craters.

Our slogan relates to how we are exploring unknown facts about the ice in the bottom of the craters.

"Ice Breakers: Cracking the Moon's Cold Secrets."

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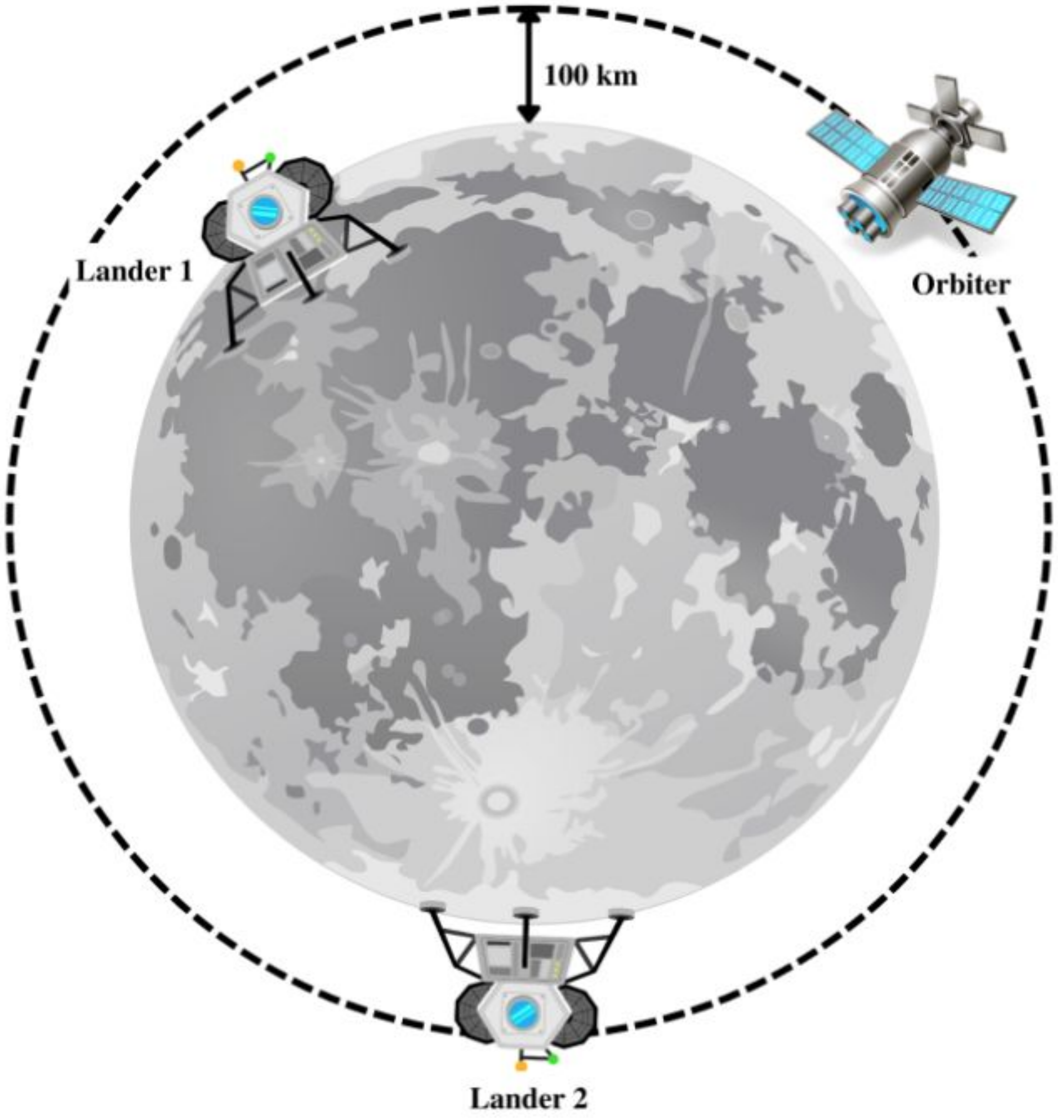
UAH Baseline Mission

- **Orbiter (Vader)**
  - 100 km circular polar orbit
  - Two Earth-years duration (24 months)
  - Communications relay to Earth
- **Lander 1 (Luke)**
  - Lands at Aristarchus crater in the Aristarchus plateau located in the northern hemisphere
  - 1 Earth year duration (12 months)
- **Lander 2 (Leia)**
  - Lands at Shackleton crater in the Aitken-Basin located in the southern hemisphere
  - 1 Earth year duration (12 months)

Mission Timeline	Year 1											
	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Lunar Oribter												
Lanar Lander 1												
Lunar Lander 2												

Mission Timeline	Year 2											
	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Lunar Oribter												
Lanar Lander 1												
Lunar Lander 2												



Selection of
Science Objectives

Potential Science Objectives

Look for biominerals delivered from asteroids

- find biominerals on the surface of craters
- finding biominerals would help prove life outside of earth

Research lunar ice that develops inside the bottoms of cold craters

- research the purity of ice in the craters
- astronauts could use ice to reduce costs for space delivery significantly

Look for evidence of lunar volatiles inside of asteroids

- explore craters for asteroid fragments or evidence of explosions
- would help us better understand the early solar system/galaxy

Science Objective Trade Study

Figures of Merit (FOMs)	Weight	Biominerals		Lunar Ice		Explosive residues	
		Raw Score	Weighted	Raw Score	Weighted	Raw Score	Weighted
Interest of Team	9	3	27	3	27	1	9
Applicability to other science fields (broadness)	1	9	9	3	3	3	3
Mission Enhancement	1	9	9	9	9	9	9
Measurement Method (easy to obtain)	9	1	9	9	81	3	27
Understood by the Public	9	3	27	3	27	3	27
Creates excitement in the public ("wow factor")	3	9	27	3	9	3	9
Ramification of the answer	3	3	9	3	9	3	9
Justifiability (nice, neat package), (self-consistent)	1	3	3	9	9	3	3
TOTAL			120		173		96

Science Objective Chosen

Research and analyze lunar ice that develops inside the bottom of cold craters.

Explanation: We find information in the bottom of these craters that would tell us how pure the ice is, and if it could be usable for the astronauts.

Importance: The ice can make it easier to travel to the Moon, and it would reduce costs significantly, since it takes a large amount of money to send supplies to the Moon.

Science Traceability Matrix

Science Objective	Measurement Objective	Measurement Requirement	Instrument Selected
Analyze lunar ice developed in the bottom of craters	Ice composition	- Measure at least 3 feet below the surface - Operate at temperatures around -388° F	Ground-Penetrating Radar (GPR)
	Ice purity	- Analyze the composition of the ice - Operate without gravity	Optical Spectrometer
	Ice temperature over time	- Measure a wide range of temperatures - Record how the temperature changes over time	Thermocouple
	Track position throughout the mission	- Tracks position - Sends position to desired location	Inertial Measurement Unit (IMU)

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Instrument Requirements

Instrument	Mass (kg)	Power (W)	Data Rate (Mbps)	Dimensions (cm)	Lifetime	Frequency	Duration
Mini GPR	0.05	2	2	5 × 10 × 2	48 hrs	30 min	10 min
Miniature Optical spectrometer	0.76	0.7	0.048	2.78 × 2.5 × .39	48 hrs	continuous	continuous
Inertial Measurement Unit (IMU)	0.55	2	5.18	3.9 x 4.5 x 2.2	48 hrs	continuous	continuous
Thermocouple	0.020 / meter	N/A	$1.0 \times 10^{(-4)}$	Wire (length TBD)	48 hrs	continuous	continuous

Support Equipment Requirements

Component	Mass (kg)	Power (W)	Data Rate	Dimensions (cm)	Other Technical Specifications
Transceiver	0.15	0.0252	5 Mbps	0.673 x 2.55 x 0.279	N/A
Batteries	1.44	400 Whr/kg 642.4Whr	N/A	8x4x8	N/A
On board computer	0.31	0.4428	10 GB onboard storage	9.5 x 8.9 x 2.32	2 MB Program Memory Size
Antenna	0.1 kg	0.02	5 Mbps	0.98	N/A
Solar panels	0.125	N/A	N/A	16x16x0.5	20% efficiency, 43.2 Whrs

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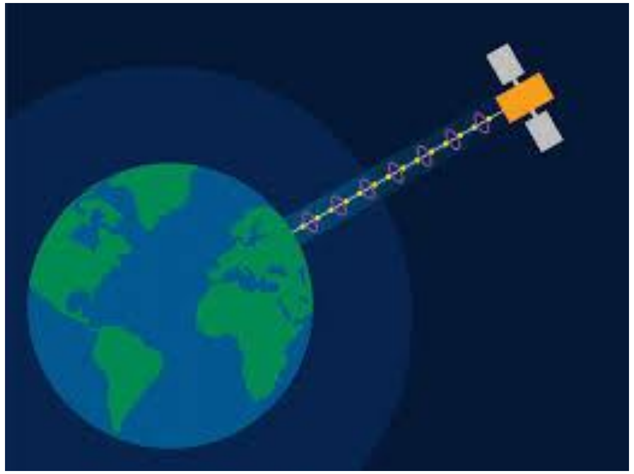
Payload Design Requirements

Project Requirements:

- Research is conducted away from the spacecraft
- Mass of 10 kg or less
- Volume of no more than 44 x 24 x 28 cm
- Poses no harm to the UAH-designed spacecraft
- Will survive all external environments during duration of experiment

Functional Requirements:

- Deploy
- Provide its own power
- Able to send data
- Able to collect data
- Take measurements
- House payload



Payload Destination

- Our payload will be landing on the Aristarchus Crater on the southern pole of the moon.

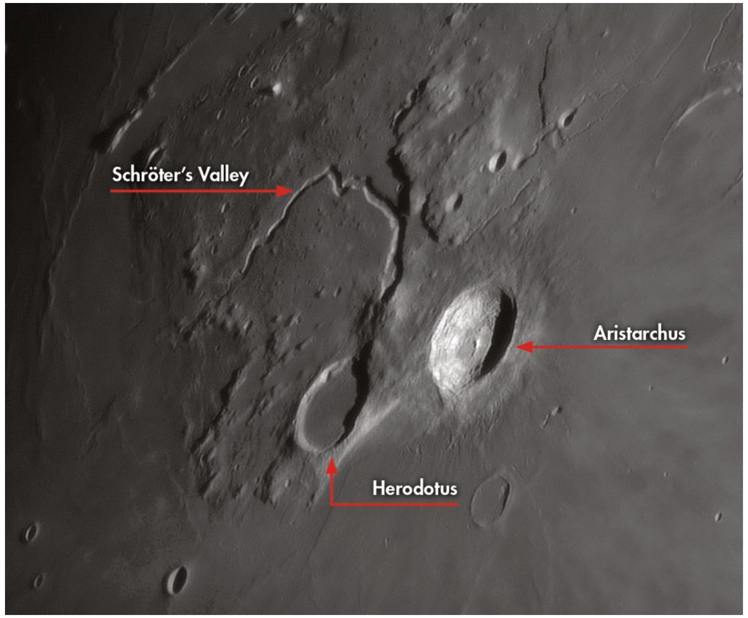
- Temperature
  - Minimum: -414.4°F
  - Maximum: 253.4°F

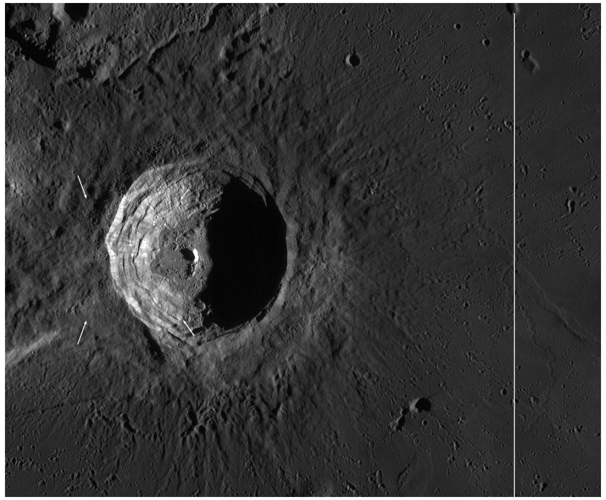
- Pressure: Zero pressure, almost a perfect vacuum

- Gravity: 1.62 m/s²

- Radiation: High-energy particles
  - Galactic Cosmic Rays, Solar Energetic Particles

- Chemistry: Oxygen-rich lunar rock with shadowed regions that contain Hydrogen, Carbon Dioxide, and Ammonia





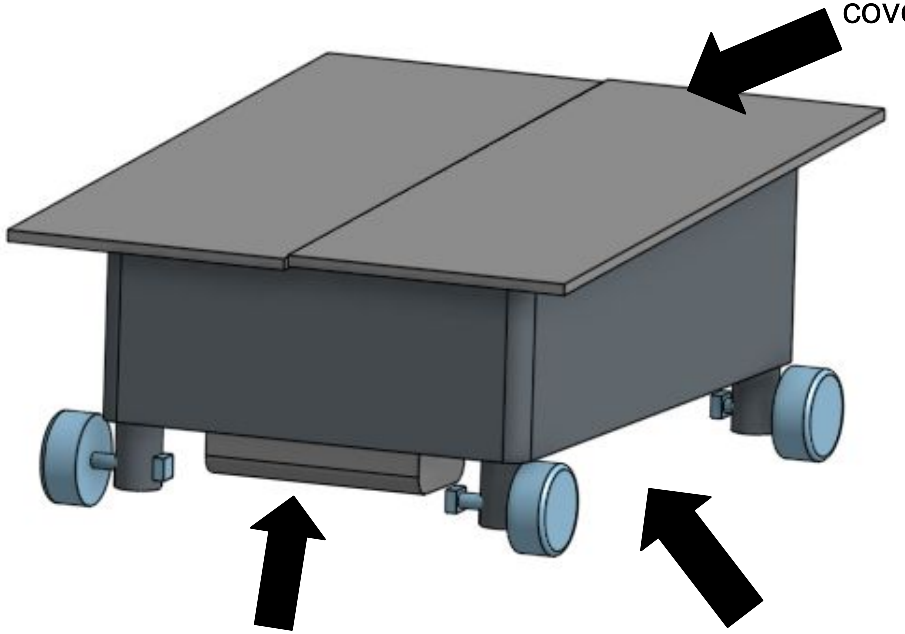
Selection of

Payload Concepts

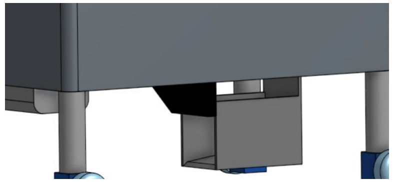
These concepts were created before the addition of solar panels in the final design

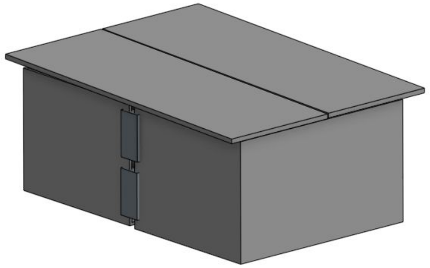
Fortitude

- Fortitude is a rover with a more durable design, made to tank any damage that’s given to it.
- Most main components are stored inside.
- This can handle the terrain in the crater.
- It’s stored in its box with a seam meant to break around each half of the box for early deployment.
- Dimensions:32x13x24cm



Scoop underneath

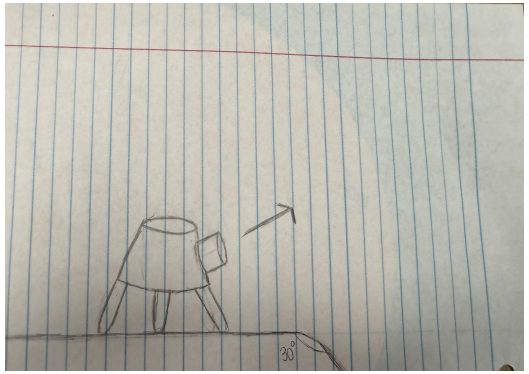




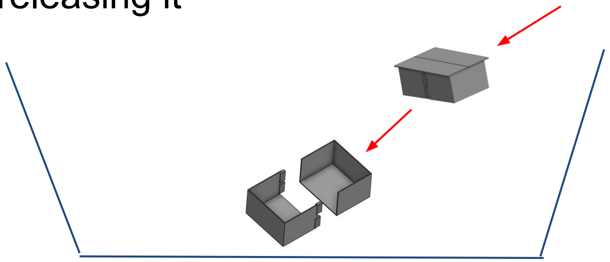
(payload cased for launch)

Fortitude - Concept of Operations

Step 1: Payload is launched into Aristarchus crater from Lander 2

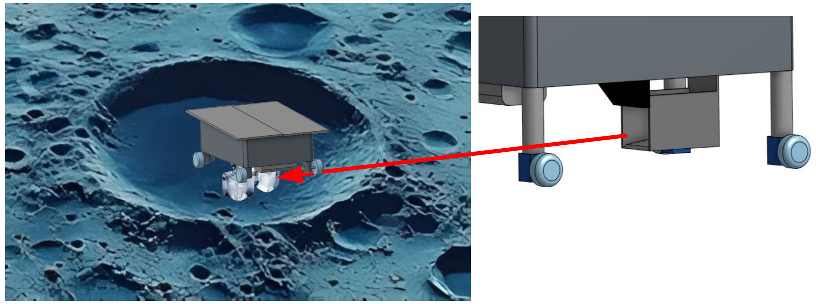


Step 2: Payload lands in the crater, causing its container to break, releasing it

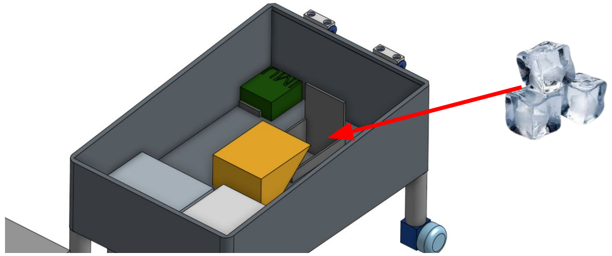


(Crater)

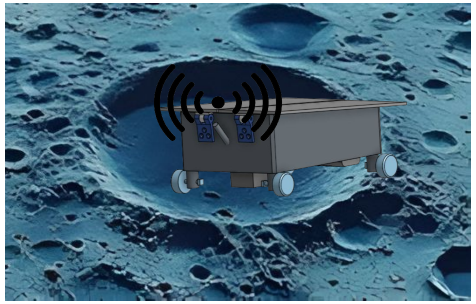
Step 3: The payload searches around the crater for ice and drives over it, scooping it up



Step 4: Payload scans the ice from an interior compartment

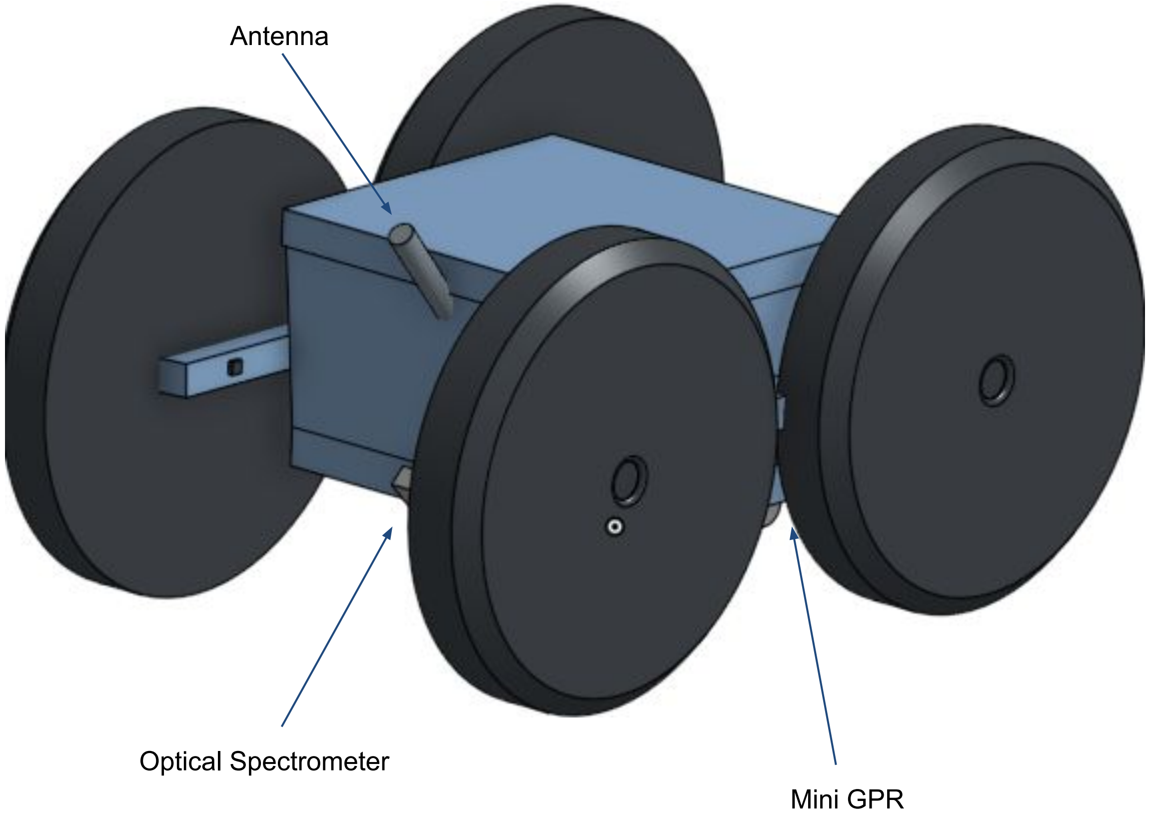


Step 5: Payload transmits information learned back to NASA



Agility

- Smaller, more power efficient design, that is light and cheaper, but it's less adapted to be durable prioritizing lower weight and frame
- Can go on very steep surfaces and is safe to flip because of its tall wheels
- Simple design, key components are stored inside the main compartment
- Contains an optical spectrometer, miniature ground penetrating radar, and wider and longer frame
- 25x15x19cm



Concept Selection Trade Study

Figures of Merit (FOMs)	Weight*	Fortitude		Agility	
		Raw Score	Weighted	Raw Score	Weighted
Science Objective	9	9	81	9	81
Likelihood Project Requirement	9	9	27	9	27
Science Mass Ratio	3	3	9	9	27
Design Complexity	3	3	9	3	9
ConOps Complexity	9	1	9	3	27
Likelihood Mission Success	9	3	27	3	27
Manufacturability	3	3	9	9	27
Cost*	1	3	3	9	9
Durability*	9	9	81	3	27
Team preference*	1	3	3	3	3
TOTAL			258		285

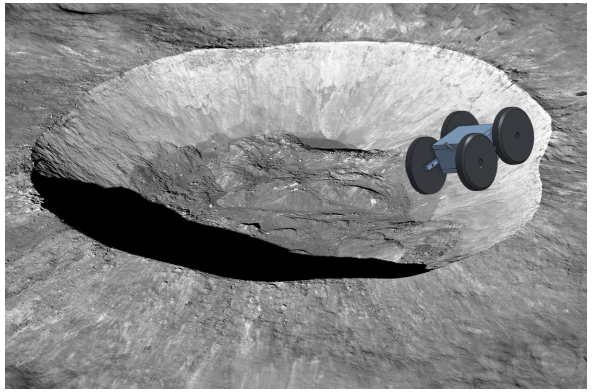
Material Selection Trade Study

FOMs	Weight	AA2024-T4 Aluminum		VIMI 115 Titanium		A286 Stainless Steel	
		Raw Score	Weighted	Raw Score	Weighted	Raw Score	Weighted
Density	9	9	81	3	27	1	9
Cost	1	9	9	1	1	3	3
Manufacturability	3	9	27	1	3	3	9
Durability	9	3	27	9	81	9	81
TOTAL			144		112		102

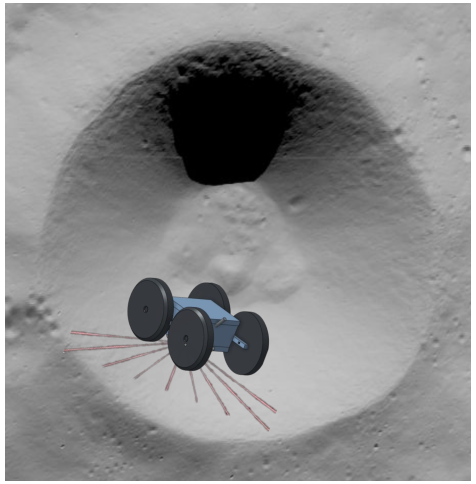
7.29kg total material weight on rover
(including carbon fiber wheels and motors)

Agility - Concept of Operations

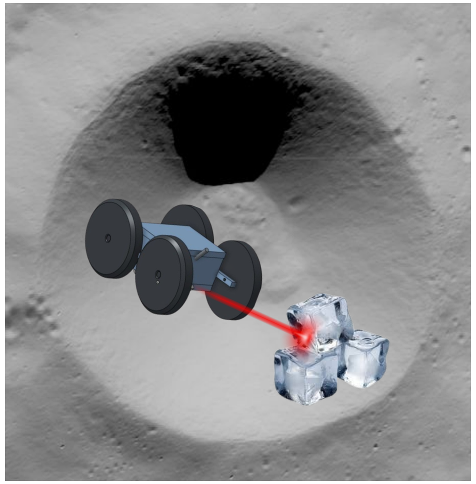
Step 1: Payload drives down into Aristarchus crater



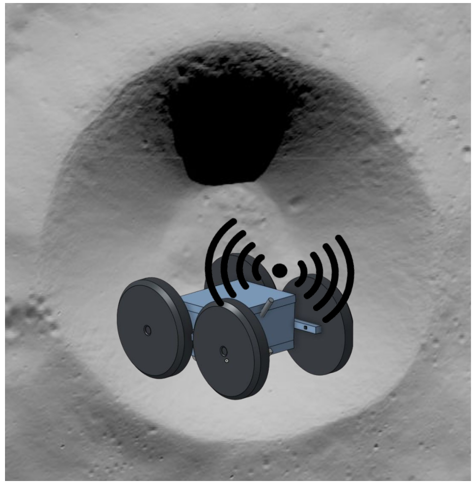
Step 2: Payload begins searching for ice inside of the crater



Step 3: Payload analyzes the ice to view it’s purity

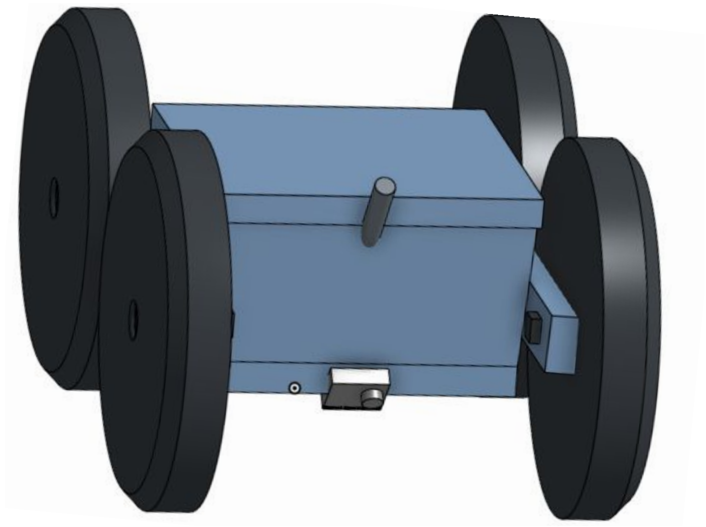


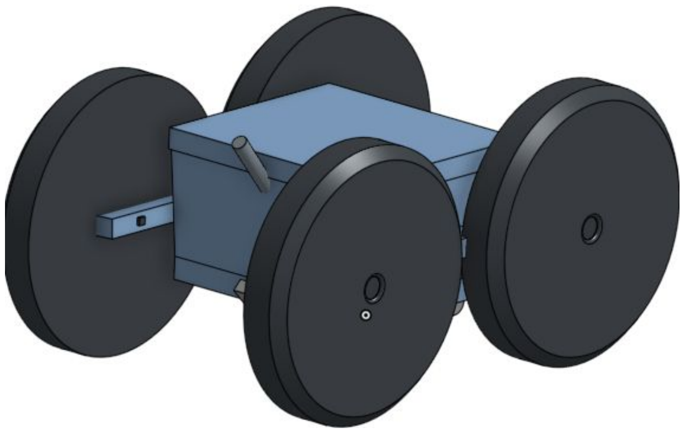
Step 4: Payload transmits information learned back to NASA using antenna



Chosen Concept: Agility

- Agility is very light, can handle bumps due to it's big wheels, and is less complex
- The cons of this concept are that it wouldn't be able to examine ice in as much detail, and it is less durable
- Made the middle axl 100% wider for added durability and stability for the wheels to mount onto, and the wheels itself were made bigger for more stability





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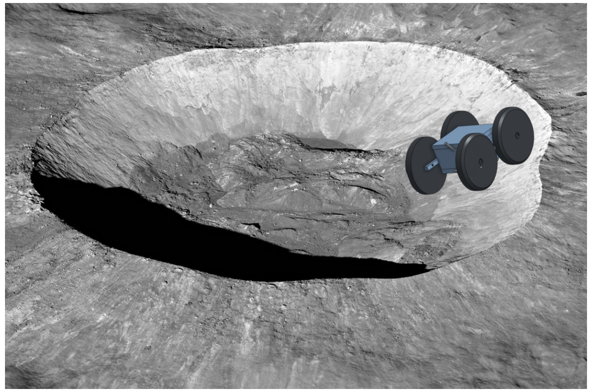
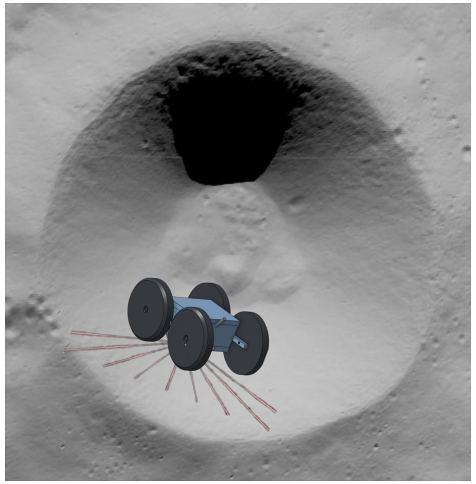
Winner’s Concept of Operations

Step 1: Payload drives down into Aristarchus crater



Step 2: Payload begins searching for ice inside of the crater



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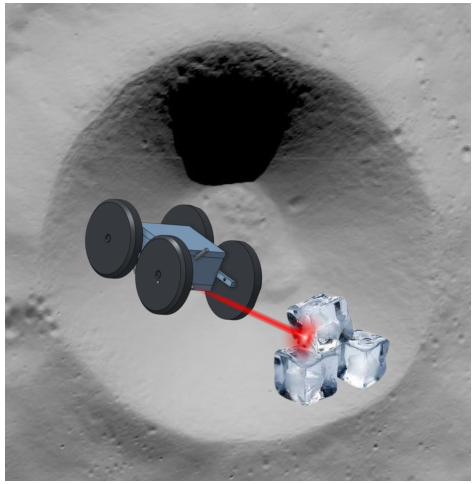

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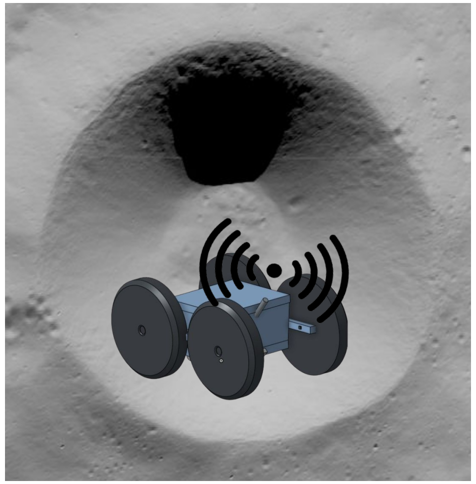
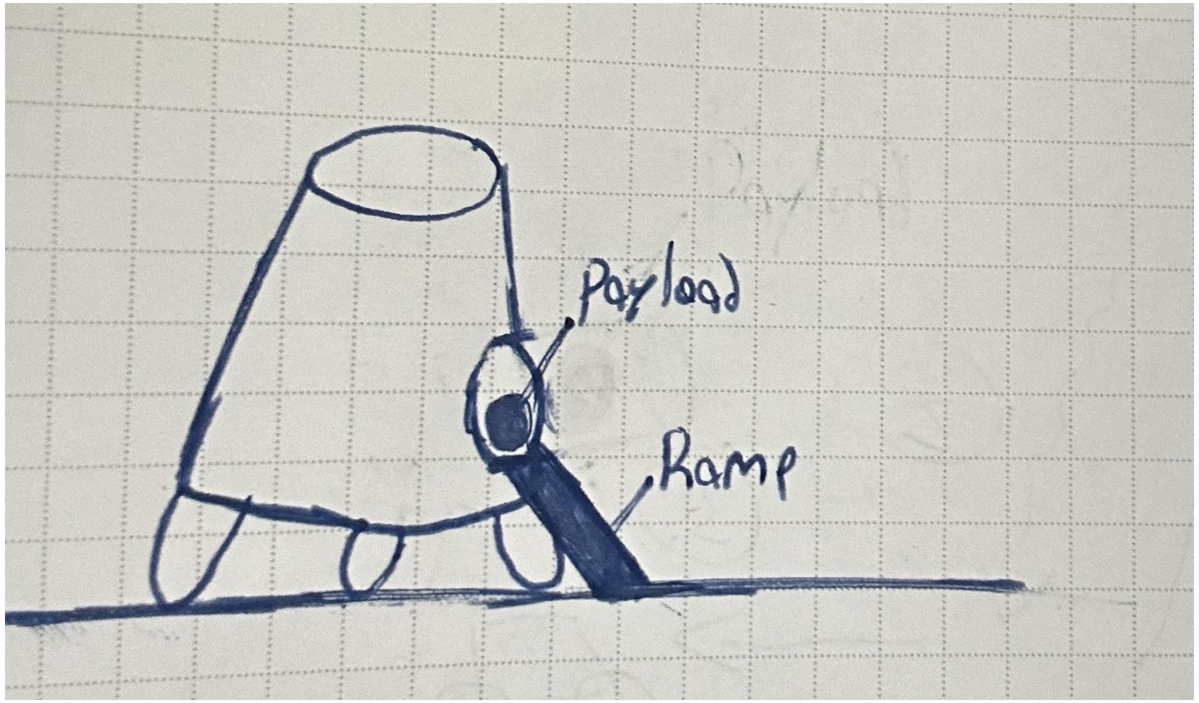


Engineering Analysis

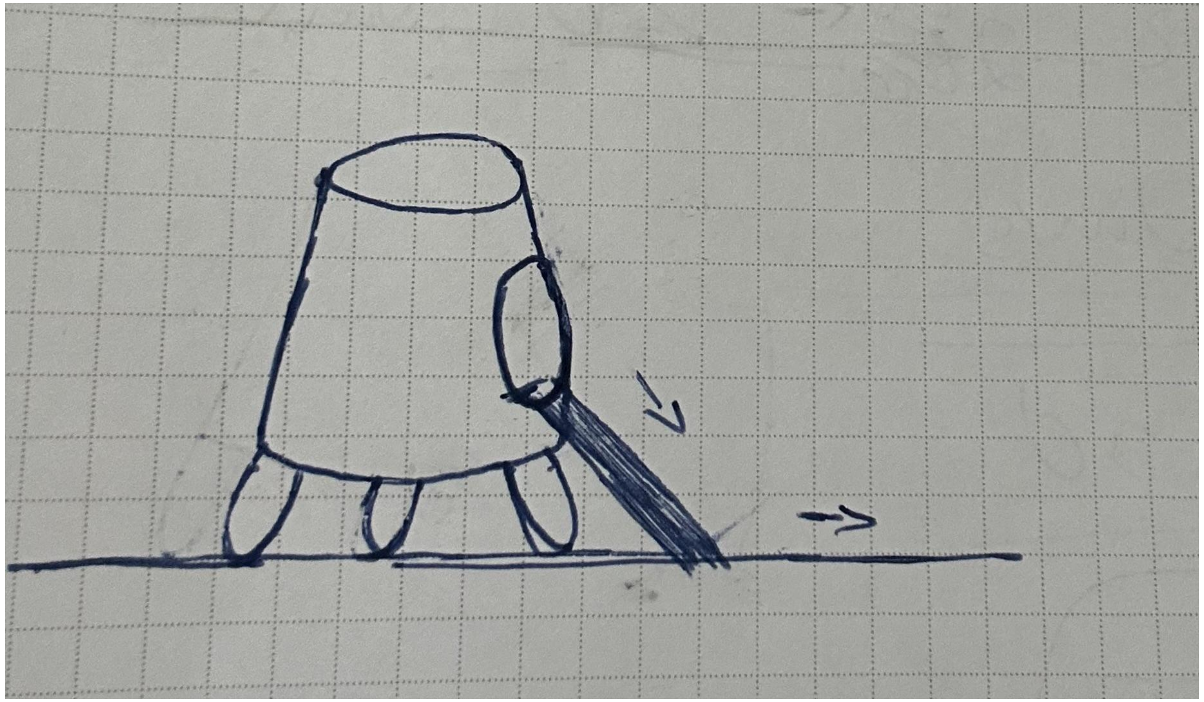
Analysis: Initial Conditions

- Given:
  - At rest
  - Gravity - 1.62m/s
- Assumptions:
  - Available ramp
  - Lander inclination ≤ 12°
- Solution:
  - V = 0m/s



Analysis: Deployment

- Given:
  - Height = 1m
  - Mass - 9.869kg
  - Dimensions - 25x15x19cm
- Assumptions:
  - Ramp length = 4ft (1.22m)
  - Still ramp
  - Constant V = 0.8m/s
- Solution:
  - Angle = 14.48°

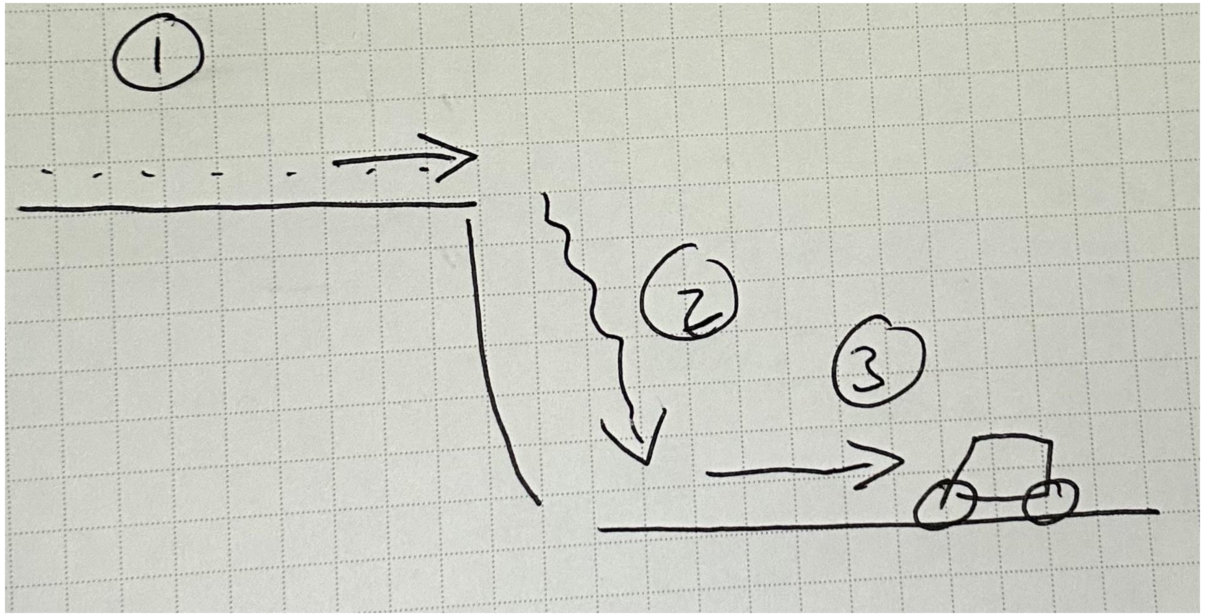


Analysis: Trajectory

- Given:
  - Gravity - 1.62m/s²
  - Crater depth - 2700m
  - Mass - 9.869 (Kg)
- Assumptions:
  - Moving at 0.8 m/s
  - Constant velocity for phase 1 (V)
  - No stops
- Solution(s):
  - Time (1) = 1250s (To crater)

  - Time (2) = 57.7s (Falling)
  - V = 93.5m/s (Final velocity)

  - Time (3) = 12.17s
  - V = 9.86m/s (Decelerating)



Analysis: Ending Conditions

Givens	Assumptions	Solutions
– Height of ramp 1m – Rover Dimensions 25x15x19cm – Gravity 1.62m/s – Rover mass 9.869kg – Surface area of crater is 296.61km²	– Ramp length 4ft – Still ramp – Moving at 0.8 m/s – Constant velocity – Stopping ~⅓ of the time	– Ramp Angle = 14.48° – Time (2) = 57.7s (Falling) – V when it hits the ground = 93.5m/s – V = 9.86m/s – time to travel the whole crater is ~74 days

Formulas:

Tanθ = opposite length/adjacent length

t=d/s (time=distance/speed)

$Vf^2 = Vi^2 + 2AD$

Vf= √2d/2

T=D/V

Analysis: Batteries

Instrument/Support Equipment	Power (W)	Operational Time (hours)	Total Power Required (W-hr)
1 - JPL mini GPR:	2	12	32
2 - Mini Optical spectrometer:	0.7	38	26.6
3 - Inertial Measurement Unit:	2	48	96
4 - Transceiver:	0.0252	48	1.2096
5 - On board computer:	0.4428	48	21.2544
6 - Motor x4	20	19	400
GRAND TOTAL:			**576**

Battery Mass =1.44kg

400Whr/Kg

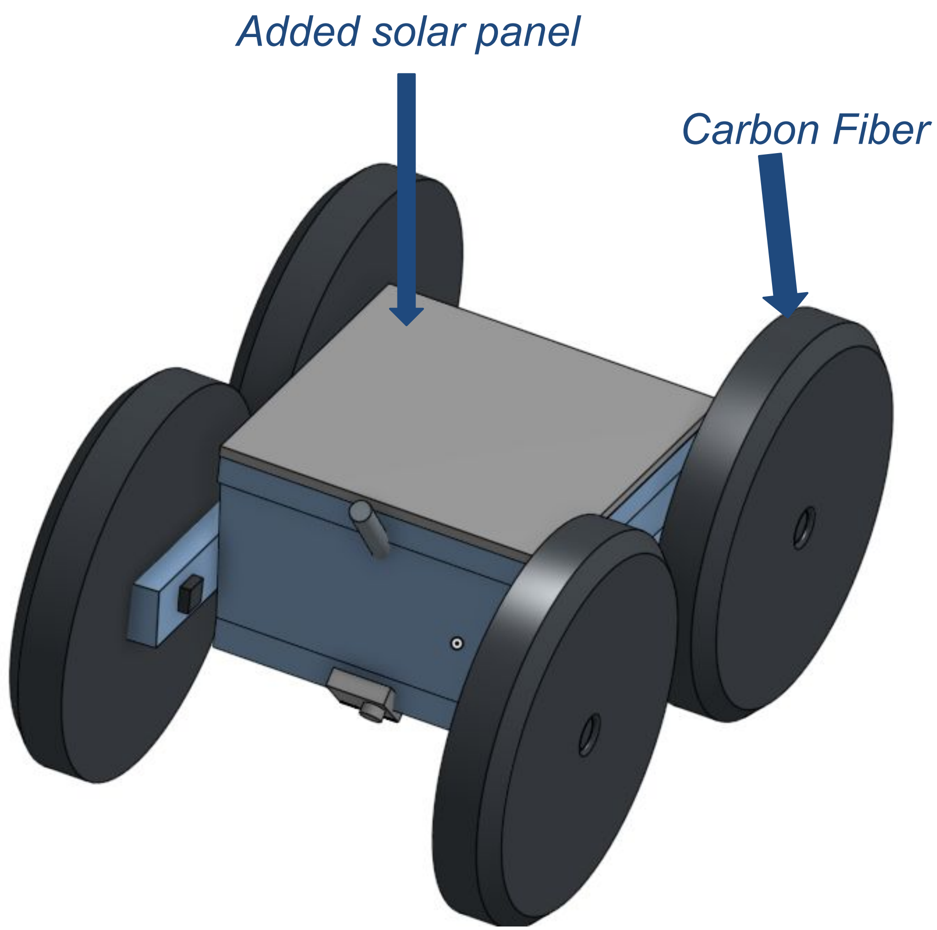
Final Design

- **Larger Wheel radius -** for our final design our design team focused on durability, the wheels have a larger radius allowing for the rover to be apart of a bigger impact with little-to-no damage; specifically to accommodate for a larger drop into the crater.

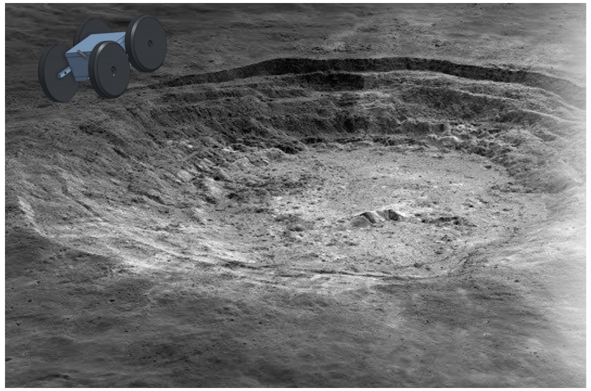
- **Lighter Wheels -** The rover's wheels are made of a carbon fiber material to keep our design light and durable.

- **Lighter frame -** This rover design has a lower total mass due to its thinner and simpler design.

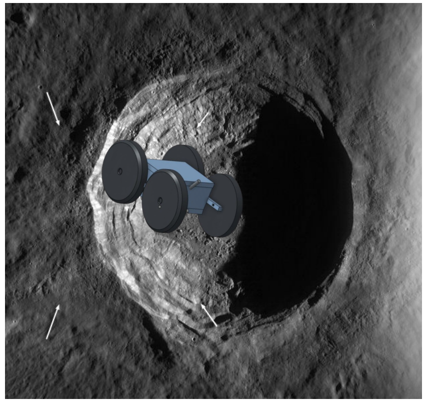
- **Solar panels for power -** The final design contains solar panels that will be used to charge the rover during the sunlight hours of our mission if they're working.



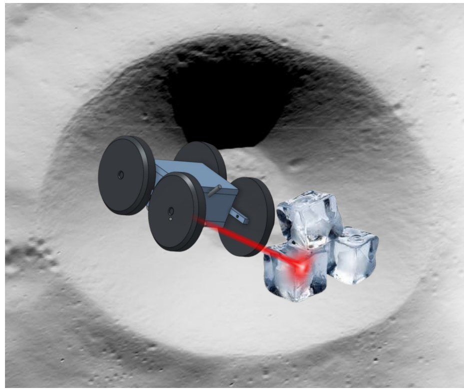
ICE Breakers' Mission



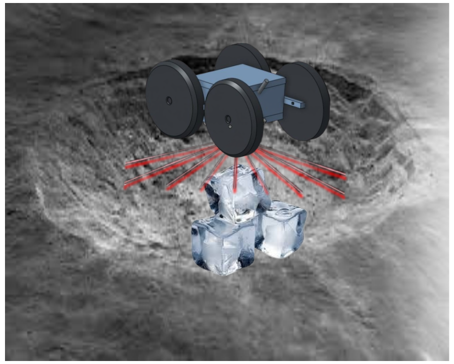
Phase I:
Payload is delivered near the Aristarchus crater.



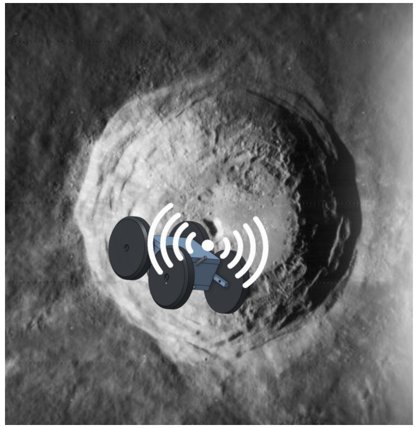
Phase II:
The payload will begin descent into crater and hits the ground at 95.5



Phase III:
The payload begins it's search for ice samples at a speed of 0.8 m/s



Phase IV:
Post ice collection, payload analyzes data.



Phase V:
Payload collects data package and reports info back to NASA via antenna

Phases 3-5 repeat until instructed otherwise

Mass Table by Function

Function	Components	Mass (kg)
Deploy	Wheels and Motors	5.494
Measure	Mini GPR+IMU	0.6
Collect Data	On board Computer	0.31
Provide Power	Batteries+Solar panel	1.565
Send Data	Antenna	0.1
House/Contain Payload	Aluminum Body	1.8
Totals:		9.869

Requirements Compliance

Requirement	Verification	Compliance
• Mass ≤ 10 kg • volume ≤ 44 x 24 x 28 cm	9.869 kg 25x15x19cm	✅

• Survive the environment
  ○ radiation
  ○ gravity
  ○ temperature
	AA2024-T4 Aluminum has been used in spacecraft framework
	The design could land any direction and not break
	AA2024-T4 has been rated for use down to -319°F
		✅

• Deploy • Measure • Collect Data • Provide Power • Send Data • House Payload	Wheels and Motors Mini GPR On Board Computer Batteries Antenna Aluminum Body	✅

Risk Analysis

Potential risks:

A) Payload gets stuck

B) Communication failure

C) Payload loses power

D) Instrument failure

E) Dust builds up & jams wheels

F) Solar panels stop working

	Low-Impact	Medium-Impact	High-Impact
High Likelihood			
Medium Likelihood	F		D
Low Likelihood		C,E	A,B

Outreach Summary

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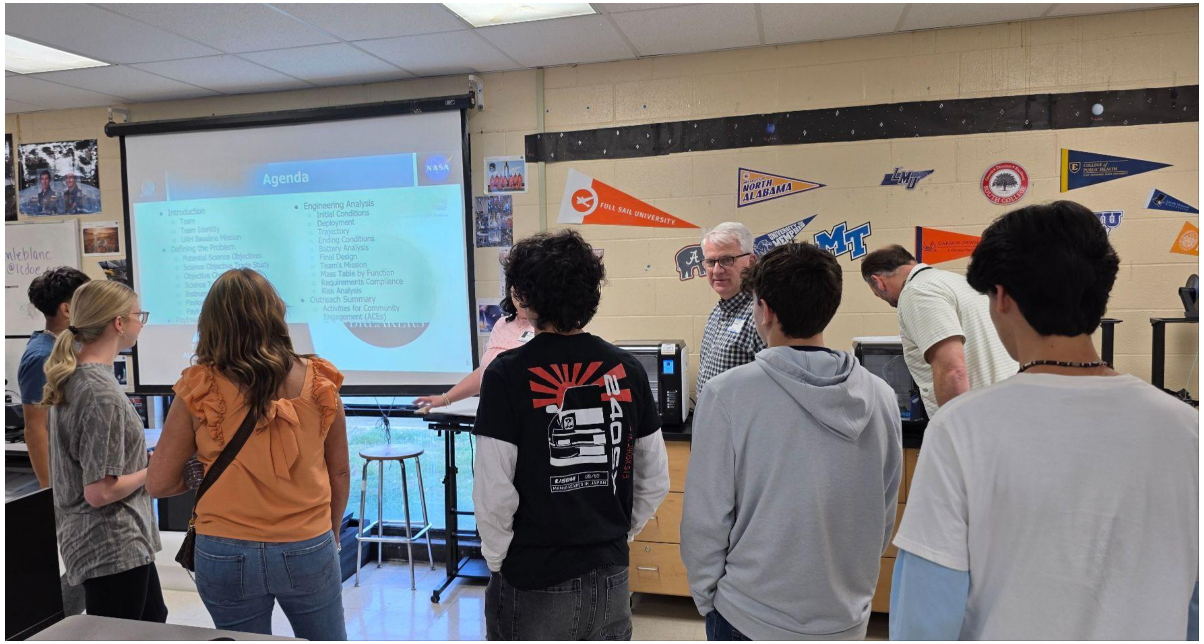
ACE #1

- ACE project #1 - February 5 at the Lincoln County High School STEM night
- We used 3D printed components to let participants construct a model of a CubeSAT satellite.
- Our mission is communicated through this project through creating curiosity about the elements of CAD and helping people understand the processes used to make our mission’s rover.

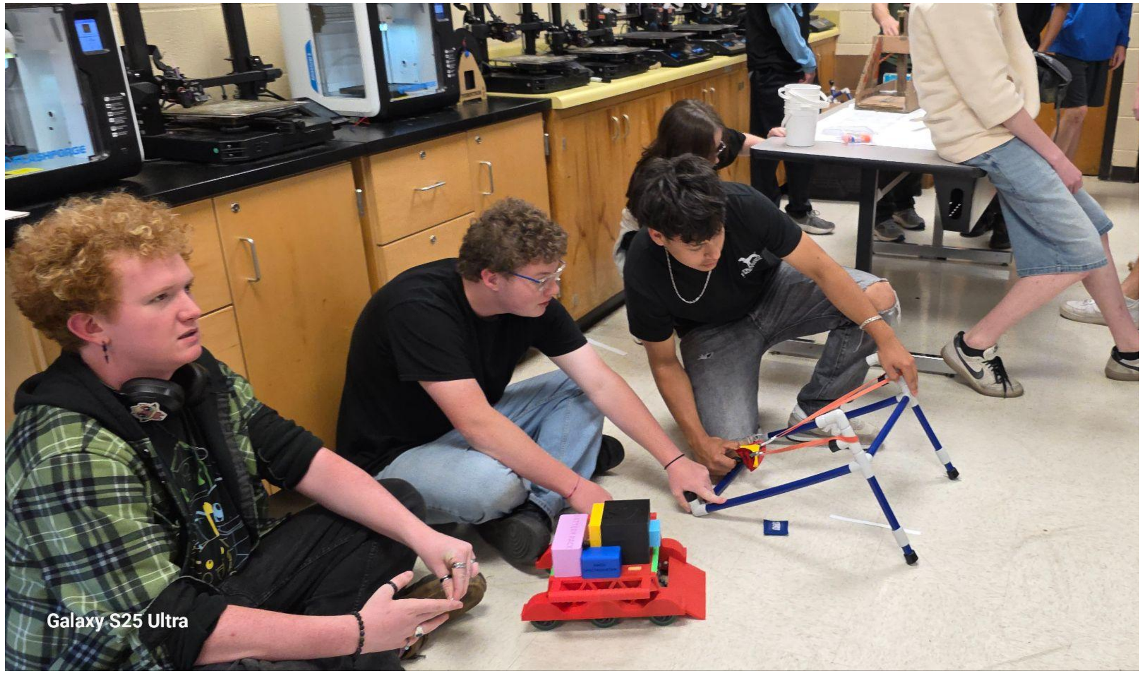
ACE #2

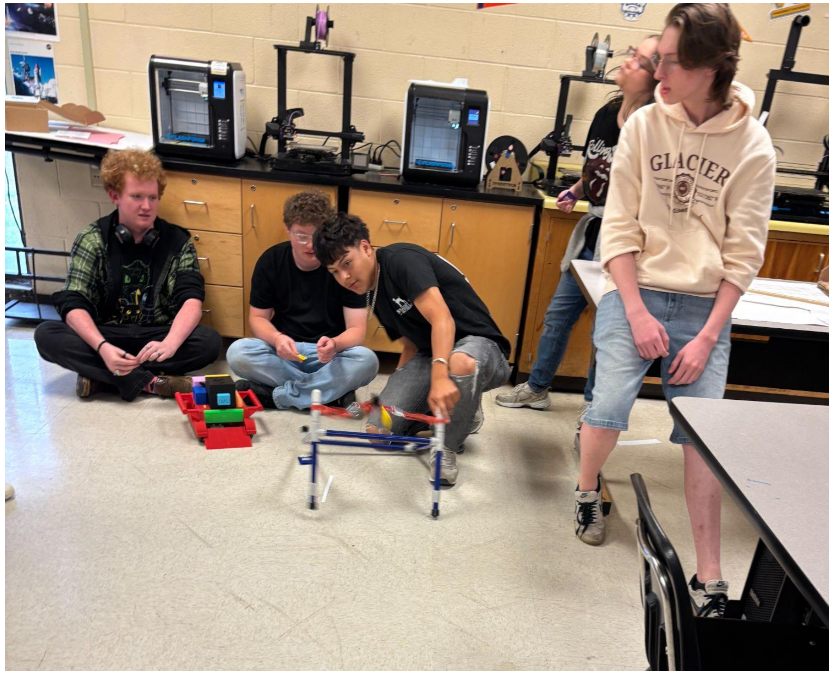
- ACE project #2 - April 1, 2026
- We communicated the importance of our mission and payload to the Kiwanis Club.



ACE #3

- ACE project #3 - April 13, 2026
- We simulated launching our payload into the crater with classes from our school.
- Our mission was communicated through giving our peers knowledge about how our payload will be drawn into the crater.





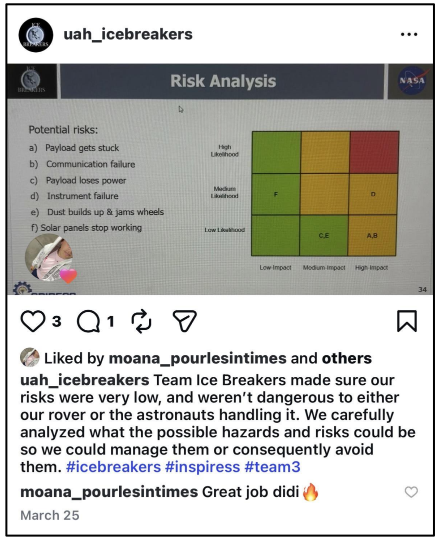
STEAM Night

- April 17, 2026
- We had over 500 people participate in our STEAM Night.
- We organized and led hands-on activities that helped give our community more knowledge about STEM. https://www.facebook.com/lcdoe.org

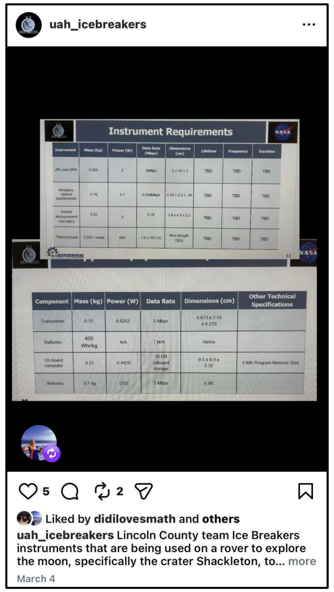


Outreach - Instagram

@UAH_ICEBREAKERS



- Our Instagram account can be found @uah_icebreakers
- We posted every Friday, alternating posts with required and non-required content.
- Our target audience for this app is teenage and older groups.
- We posted information about our research, and team members.



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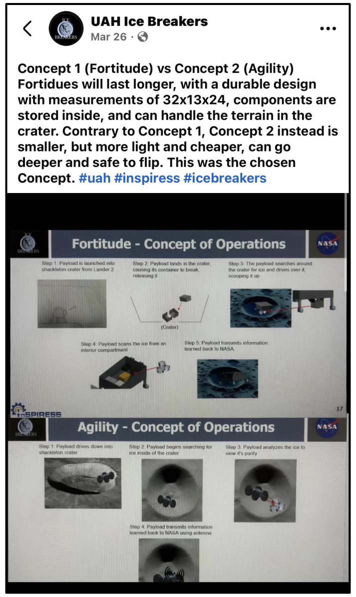
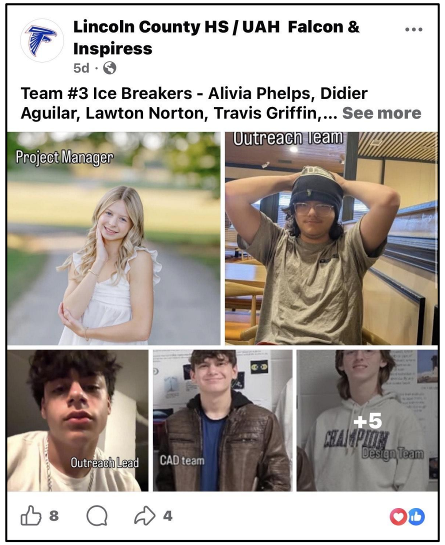
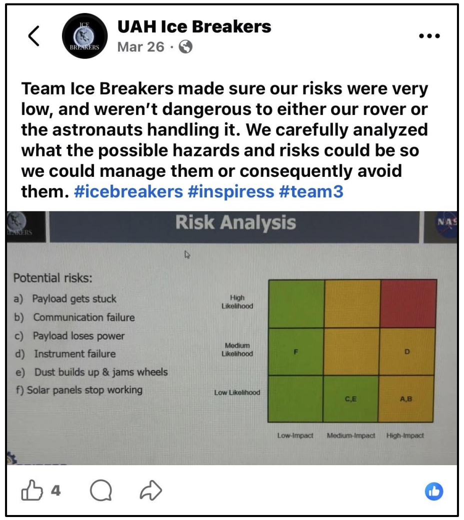
Outreach - Facebook



- Our Facebook account can be found @Lincoln County HS/ UAH Falcon & INSPIRESS
- We posted every Friday, alternating posts with required and non-required content.
- The target audience for this app is older groups and parents.
- We posted information about both our research and goals.





Outreach Analytics

Likes - 22	Views - 121	Followers - 10

Likes - 24	Views - 144	Followers - 3

Closing Remarks

ICE BREAKERS

Lessons Learned

Leadership

- Skills learned: teamwork, problem solving skills, adaptability, & creative thinking
- What was most unexpected was that we would be calculating trajectory and movements of our rover
- Something we would work on would be better communication between ALL team members

Design

- We learned how to 3D model payloads effectively while making sure everything worked together
- The hardest problem within this project was making sure our concept would survive landing into the crater
- We would put more detail into the modeling of the individual parts

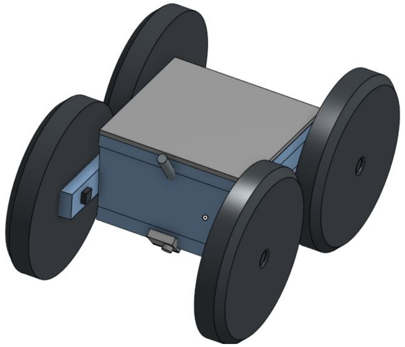
Outreach

- We learned how to communicate with the public effectively
- The most difficult part of this project was planning the ACE events in places we could reach everyone
- We would put more time into posts and coming up with fun events for the public

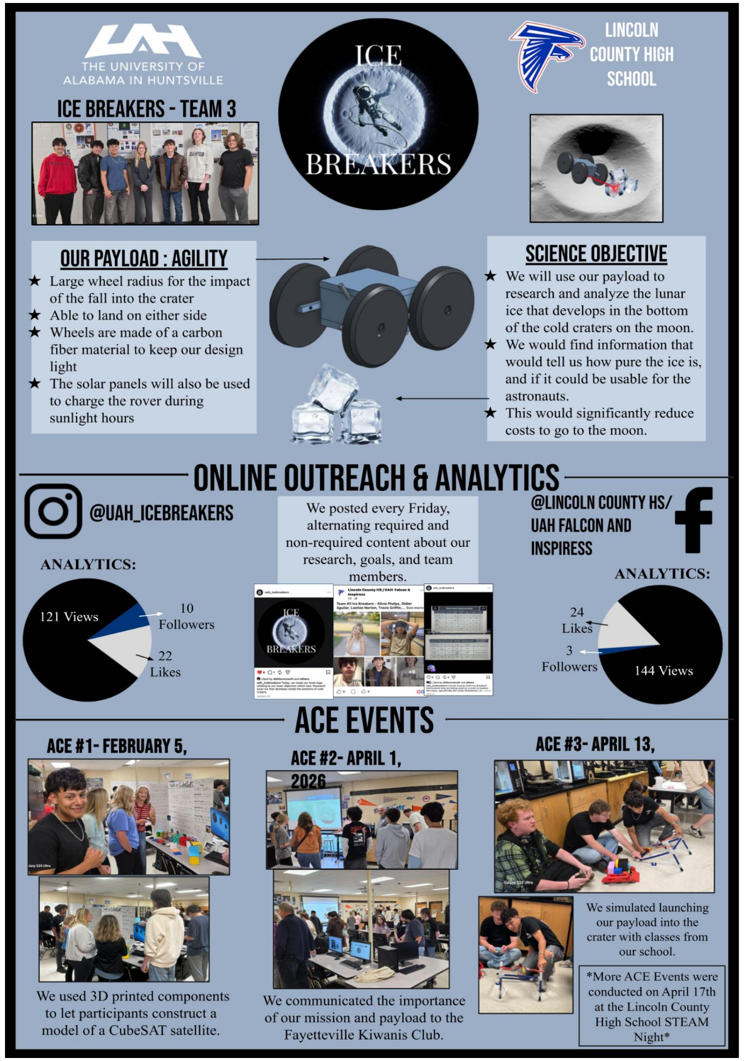
Summary

AGILITY:

Agility is a small, simple, power efficient rover that is designed to research and analyze the purity of ice using it's various instruments, as well as land on it's wheels no matter how it falls or tumbles.



Science Objective:
Research and analyze lunar ice that develops in the bottom of craters.



OUTREACH:

@UAH_ICEBREAKERS

Analytics:	Analytics:
Likes - 24	Likes - 22
Views - 144	Views - 121
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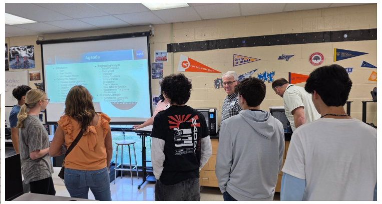
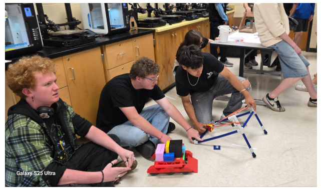
ACES:





inSPIRESS
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Sources

- Google (https://www.google.com)
- Nasa websites (https://science.nasa.gov/moon/facts/)
- UAH Notebook (https://uah.instructure.com/courses/71768/modules)
- Peers with similar objectives/processes as us

Thank you all for listening to us!

ICE BREAKERS